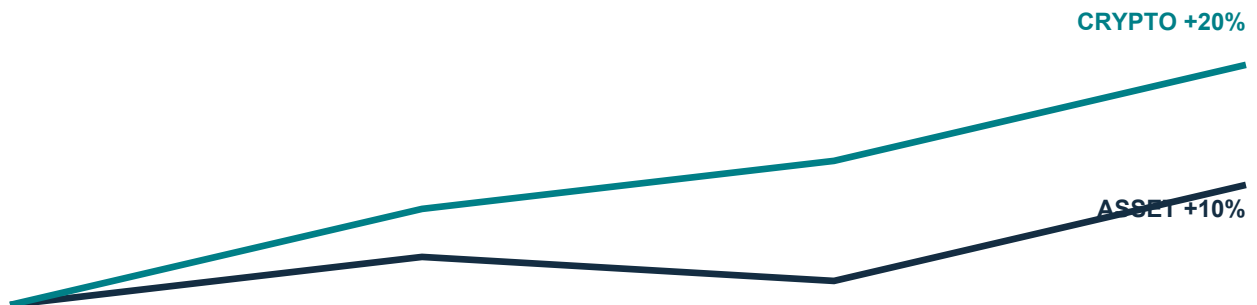


COX

Crypto Outlook Index

Measuring and trading crypto
price performance relative to CRYPTO

PRICE-ONLY / ONE-MINUTE PUBLICATION / CLOSED COLLATERAL



Illustrative normalized paths. Asset rises; its COX reference falls to 91.67.

Version 0.1 | Draft for review

10 October 2026

A standalone product paper derived from the Relative Performance Market framework. Financial mechanism proofs and production security remain engineering deliverables.

Contents

A guide to the product, the reference mathematics and the financial boundaries.

Executive summary	3
1. The product proposition	4
2. Product commitments and design boundaries	5
3. The reference mathematics	6
4. What CRYPTO represents	7
5. Shared comparisons and the role of CRYPTO	8
6. Evidence, quality and verification	9
7. The one-minute publication cycle	10
8. Binding requests and the timing boundary	11
9. The closed collateral system	12
10. Why a COX reference is not a payout formula	13
11. Continuous entry, switching and redemption	14
12. A worked publication and user journey	15
13. Product surfaces, candles and expectations	16
14. System architecture and persistent state	17
15. Security, failure and governance	18
16. What changes from EOX Paper 3	19
17. Validation program and remaining decisions	20
Appendix A. Canonical terminology	21
References	22

Executive summary

COX means Crypto Outlook Index. Its proposition is straightforward: a crypto asset can rise in price while underperforming the crypto market, or fall while outperforming it. COX makes that comparative outcome the reference for a continuously accessible market.

Performance means price performance. COX does not score network adoption, developer activity, transaction volume or protocol fundamentals. Market capitalization and volume do not contribute to the performance score. An accepted reference price, measured against the same period's CRYPTO benchmark, determines the relative reference.

The proposed product has two connected engines. The reference engine calculates what happened to asset prices relative to CRYPTO. The collateral engine determines how existing backing is redistributed among participating claims. Keeping those engines distinct is essential: an asset's relative return cannot automatically be promised as the same percentage return on every user's position in an unevenly allocated closed pool.

The operating cadence is one minute. Users may submit deposits, allocation changes and redemption requests continuously. Requests admitted to a batch execute at its next valid publication after existing positions have been revalued. A new entrant cannot purchase performance that occurred before entry, and a departing user cannot redeem against a stale value that avoids an already-observed loss.

The pool creates no money to fund winners. With external cash flows and fees accounted for separately, one claim's gain is funded by reductions in other claims. There is no borrowing, leveraged exposure or required terminal maturity in the proposed product. Continuous redemption means access through the publication process, not guaranteed instantaneous execution during an oracle or network failure.

Evidence remains essential. Confidence becomes a price-quality admission function rather than a multiplier on performance. The initial architecture omits a dedicated optimistic challenge window around every publication and relies instead on authenticated inputs and deterministic verification. That choice remains conditional on an adequate source and verification design.

The paper preserves the central RPM question: how can comparative performance become a financial exposure? It narrows the measurement problem from heterogeneous economic statistics to prices, while making the continuous closed-pool problem more explicit. The unresolved work is precise: benchmark construction, the transfer rule, fair batch accounting, exceptional-state handling and economic security. [1, 2]

A rising price answers whether an asset gained in the quote currency. COX answers whether it gained ground against CRYPTO. A user's redeemable value is then determined by the separately specified pool mechanism.

1. The product proposition

An investor can hold a crypto asset through a rising market and still lose ground to the available crypto opportunity set. A dollar-denominated green candle alone does not resolve that comparison. COX makes the comparison explicit through a common reference called CRYPTO.

The intended user statement is: "I expect this asset to outperform CRYPTO from my entry onward." Users express that expectation by allocating collateral to a corresponding claim. They can later change allocations or redeem through the next publication process.

Asset price change	CRYPTO change	Ratio-relative change	Interpretation
+10%	+20%	-8.33%	Price rose; relative ground was lost
-10%	-20%	+12.50%	Price fell; relative ground was gained
+20%	+20%	0.00%	Asset matched CRYPTO

These examples use the ratio of gross returns. They are not percentage-point subtraction. The first row is also a 10-percentage-point shortfall, but that is a different representation of the same comparison. The paper uses ratio-relative change consistently.

COX exposure does not confer ownership of BTC, SOL or another underlying asset. Collateral backs a claim inside the COX system. The underlying assets' prices supply the reference; the protocol does not need to purchase those assets merely because users select their names. Whether the collateral itself is held on chain or through another custody arrangement is an implementation decision.

The initial discussion used BTC, SOL, ZEC, ETH and "STARK" as examples. They are not a finalized listing policy. Every admitted asset needs an unambiguous identifier and a validated price source; a ticker alone is insufficient. The intended identity behind "STARK" must be resolved before any asset is listed.

COX's distinctiveness does not rest on claiming that relative trading is new. Price ratios and portfolios already express comparative views. The research proposition is a unified experience with a shared benchmark, scheduled publication, allocation-based participation and conserved collateral. Whether that structure is more useful than alternative exposures must be demonstrated through mechanism analysis and user testing, not assumed from the name RPM. [1, section 34]

The word "outlook" describes the future view that motivates a user's allocation. The verified reference is a backward-looking calculation from accepted prices. It is not an operator's forecast.

2. Product commitments and design boundaries

The following commitments define the paper's scope. Detailed engineering choices must preserve them or explicitly propose a new product version.

Commitment	Meaning
Price-only performance	The performance variable is the asset's reference price change
Shared CRYPTO benchmark	All asset references use the same published benchmark version
One-minute publications	Each scheduled minute is an opportunity to commit a new valid state
Next-publication execution	Accepted requests execute after revaluation at the applicable publication
Closed collateral	Revaluation redistributes backing rather than manufacturing liabilities
Continuous participation	Users can request entry, switching and redemption without waiting for maturity
No leverage	No borrowing or multiplier promises amplify financial exposure

The paper uses a reference starting at 100 and sensitivity of 1x. The number 100 is a display base, not a return multiplier. Changing the display base cannot change any holder's rights.

Evidence and reproducibility remain part of the design. Price quality determines whether inputs can be admitted. A weak observation does not turn a measured 10% price change into a confidence-discounted 6% change. A missing observation does not become a zero price.

There is no separate optimistic assertion and challenge cycle in the initial COX publication path. This does not mean that data are trusted without verification or that operational disputes disappear. The system must verify the chosen source's authentication, timestamp and admissibility rules before committing a state.

Several choices remain open: CRYPTO membership and weights; the reference-price source and observation window; the treatment of a holdable CRYPTO claim; the exact transfer rule; fee and rounding rules; collateral and custody; and the treatment of prolonged failures. This paper does not select an oracle vendor, blockchain or legal structure.

The closed-pool requirement is a specification to prove. EOX Paper 3 and the RPM paper describe the same conservation ambition while leaving the complete entry, exit and payout mechanism unfinished. COX must not inherit an assertion that those problems were already solved. [1, sections 10 and 36; 2, sections 19-23]

Status convention: product commitments are intended behavior; equations marked illustrative explain a constraint; open mechanism questions remain engineering deliverables. No section describes a deployed COX system.

3. The reference mathematics

Let k denote a successful publication and to a common reference origin. $P_{i,k}$ is the accepted price of asset i in the declared quote currency. B_k is the level of CRYPTO under the active methodology. The initial domain assumes strictly positive prices and benchmark levels; terminal zero-price events require the special handling described later.

Symbol	Definition
$P_{i,k}$	Accepted reference price for asset i at publication k
$A_{i,k}$	Asset price relative to its value at the common origin
B_k	CRYPTO index level
W_k	CRYPTO level relative to its value at the common origin
$Q_{i,k}$	Asset's COX reference, initially 100

$$A_{i,k} = P_{i,k} / P_{i,0} \quad W_k = B_k / B_0$$

$$Q_{i,k} = 100 \times A_{i,k} / W_k$$

The change in the COX reference between consecutive publications is:

$$Q_{i,k} / Q_{i,k-1} - 1 = (1 + r_{i,k}) / (1 + r_{B,k}) - 1$$

Here $r_{i,k}$ is the asset's simple price return and $r_{B,k}$ is CRYPTO's simple return for the same interval. Comparing raw dollar prices would be meaningless: a high unit price does not establish stronger performance than a low unit price.

The price definition must remain consistent through token redenominations, splits and migrations. A unit conversion that leaves an owner's economic position unchanged must not produce a fictional gain or loss. The default scope is price return, excluding staking rewards, lending yield and airdrops. Adding distributions would create a different return specification and require a versioned change.

Reference origins and user entry times are different. The public chart may begin months before a user arrives. That user enters at the current executable claim valuation and does not receive historical gains merely because Q began at 100.

After a missed publication, asset endpoint comparisons span the elapsed interval. The benchmark must separately preserve its declared rebalance calendar, as section 4 explains. COX must not invent intermediate prices or repeat returns to fill missing minutes. Records expose scheduled and actual publication times.

Fixed-point precision, rounding and overflow bounds must be specified for an implementation. The displayed decimals are not permission to round monetary accounting independently. Reference arithmetic and claim arithmetic each require reproducible rules.

4. What CRYPTO represents

CRYPTO is the common price-performance benchmark for the admitted universe. It is an index, not the sum of raw token prices and not automatically total crypto market capitalization. The word "global" describes the intended reference environment; a limited pilot roster must be labeled as such rather than presented as all crypto activity.

A price-only index still needs weights. One illustrative chain-linked construction is:

$$B_k = B_{k-1} \sum_{i \in U(k-1)} w_{i,k-1} (P_{i,k} / P_{i,k-1})$$

$$w_{i,k-1} \geq 0 \quad \sum_i w_{i,k-1} = 1$$

A publication outage must not silently skip benchmark rebalances. The system must reconstruct required intermediate observations or apply a predefined rebalance-freeze policy; endpoint returns alone need not reproduce a rebalanced index.

This equation is a benchmark example, not a selected weighting policy. Weights are determined before the measured interval. Constant equal weights each minute imply a different index from holdings established at a scheduled rebalance and allowed to drift with prices. The final specification must identify which convention applies.

For the current price-only scope, fixed or scheduled equal weights are a research candidate that requires no market-cap or volume score. They are not neutral: each eligible constituent receives the same initial influence regardless of economic size. Eligibility and duplicate-exposure rules consequently matter. Creating multiple closely related assets must not become a cheap way to influence the benchmark.

Established crypto indices show that constituent eligibility, price sourcing and weighting are explicit methodology choices. They are precedent for careful specification, not evidence that a particular COX benchmark is already correct. [4]

Membership changes require continuity. Admitting a new asset or removing an old one must not mechanically create a return at the transition. New weights apply prospectively and the index is linked across the boundary. Removing a failed asset must not erase the loss that occurred before removal.

The tradable roster and benchmark roster may differ only if that distinction is explicit. If CRYPTO includes assets unavailable as COX positions, all currently tradable assets could outperform CRYPTO together. Their references may all rise, but their redeemable claims cannot all gain dollars in an unchanged closed pool without another allocation funding those gains.

The final methodology must publish the universe, unique asset identities, weight schedule, reconstitution rules, exclusions and historical versions. The benchmark cannot be changed opportunistically after observing which choice would favor existing positions.

5. Shared comparisons and the role of CRYPTO

A shared benchmark makes cross-asset measurement consistent. For assets with the same origin and reference convention:

$$Q_{i,k} / Q_{j,k} = (P_{i,k} / P_{i,0}) / (P_{j,k} / P_{j,0})$$

CRYPTO cancels in this pair comparison. The result describes normalized price performance of i against j; it is not their raw spot exchange rate. New listings with different origins require a common comparison interval before this identity can be used as a performance statement.

This is the network property developed in the RPM paper: N canonical benchmark references can support many comparative views. It does not by itself prove that all those views can be financially netted or redeemed safely. [1, sections 15 and 25; 2, section 24]

An asset included in CRYPTO partly moves its own benchmark. Large constituent weights therefore dampen its measured outperformance relative to the rest of the basket. COX should disclose contribution and concentration. A separate benchmark excluding each measured asset could reduce self-inclusion, but would abandon the single common denominator and complicate cross-reference derivation. This paper retains a shared CRYPTO design and leaves concentration treatment to the benchmark specification.

Three objects need distinct names in the product:

Object	Economic role
CRYPTO benchmark	A deterministic price index used in the reference calculation
CRYPTO position	A possible pool claim with separately defined participation rights
Collateral	The asset actually backing redemptions

Users may reasonably want to rotate from BTC exposure into a CRYPTO position. That is a desired product capability to evaluate, not a consequence of the benchmark existing. CRYPTO compared with itself has a constant relative reference of 100; that fact alone does not define the value of a holdable CRYPTO claim.

A benchmark position could participate in pool revaluation according to benchmark growth, thereby providing an economic alternative to individual asset allocations. Its exact transfer rights remain part of the mechanism. It must not be described as guaranteed cash or a risk-free reserve unless it has explicitly different, fully backed rights.

Keeping collateral outside active allocations is another possible state. Unallocated collateral cannot quietly finance active holders' gains. If it participates in redistribution, that participation must be named and accepted by its owners.

6. Evidence, quality and verification

Price is observable, but a usable price reference still requires a definition. Different venues can trade at different prices, observations arrive at different times, and an authentic source can report a market that has been manipulated. COX therefore retains evidence while narrowing what it establishes.

Each publication should bind the asset identities, accepted prices and units, observation timestamps, source identifiers and authentication material, benchmark weights, methodology version, previous publication and resulting references. When the provider supplies an aggregate feed rather than raw trades, COX's reproducibility claim begins at that accepted feed boundary; it must not pretend to possess all underlying venue data.

The quality layer determines admissibility. It checks the configured age bound, synchronization tolerance, source availability, price validity and provider uncertainty where supported. Numeric thresholds belong to the release specification and require testing. A rejected observation remains a recorded incident rather than becoming a fabricated substitute.

Pyth's integration guidance documents stale observations, uncertainty intervals and the latency between external markets and on-chain use. It also identifies the risk that a caller selects a favorable historical update. These are useful examples of integration hazards; choosing Pyth is not a decision made by this paper. [\[3\]](#)

Confidence is not financial optimism. A high-quality price observation does not establish that an asset will rise. Nor does low quality automatically reduce measured performance. The preferred initial behavior is to admit the observation, flag a non-executable indication, or withhold the affected executable publication under predetermined rules.

Removing a dedicated optimistic challenge window changes the verification path, not the requirement to verify. Authenticated input proofs and deterministic computation should be checked before state commitment. Independent monitors should be able to reproduce published calculations and identify inconsistent outputs.

If a price meets the agreed source rules but the underlying market was manipulated, a later objection cannot invent an alternative "fair" price. Protection must come from the reference construction, eligibility, exposure bounds and incident policy. The security design must consider the total extractable value across all claims influenced by a corrupted constituent.

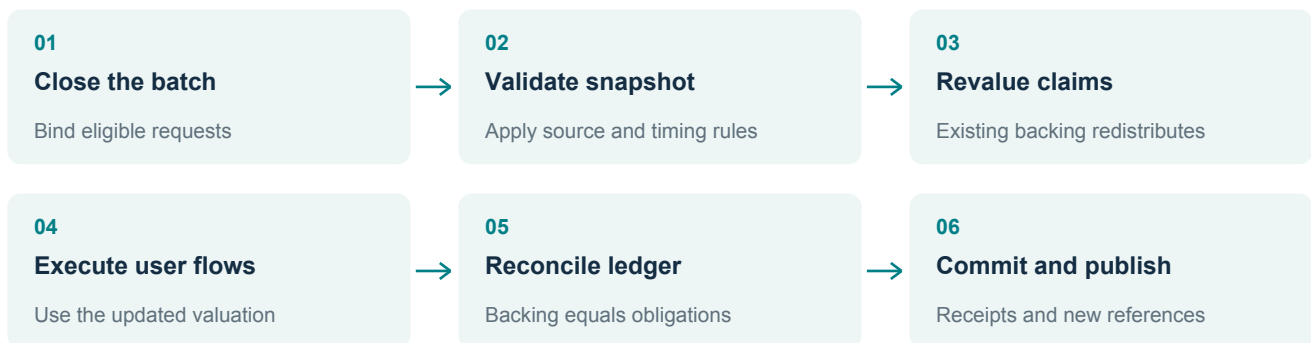
Incorrect or revised provider data require an explicit finality rule. A source correction should be preserved as new evidence. The treatment of already executed financial state must be specified in advance; an operator must not silently rewrite completed redemptions.

7. The one-minute publication cycle

COX accepts user intent continuously and processes economic state in scheduled batches. One minute is the initial product cadence, not a proven optimal parameter or a guarantee that every scheduled publication succeeds.

The central order is: close the eligible request set, validate the price snapshot, revalue existing claims, execute eligible flows at the updated valuation, then commit and publish the complete result. A request submitted after the cutoff belongs to a later batch. An acknowledgement must name the batch to which a request was admitted.

READ LEFT TO RIGHT, THEN CONTINUE ON THE SECOND ROW



The system has three relevant times. The request cutoff decides which instructions belong to the batch. The observation rule decides which prices value it. The commitment time is when the resulting state becomes authoritative. These times may be close together, but they are not interchangeable.

An illustrative schedule places cutoffs at UTC minute boundaries. The exact offset between cutoff and observation, the sampling window and the confirmation requirement remain engineering parameters. Their relationship must make inclusion and cancellation binding before the execution snapshot is known; a schedule that permits choosing participation after seeing that snapshot is inadmissible. The protocol must select observations by deterministic rules; an operator or trader cannot choose whichever acceptable-looking timestamp gives the preferred outcome.

Before revaluation, existing positions bear the entire interval since the previous accepted publication. Pending deposits do not earn that interval. After revaluation, new units are issued, redemptions burn units, and switches move current value between claim classes. The execution receipt identifies the reference and claim valuations actually used.

All batch members use a consistent valuation basis. A sequential implementation must not let the first processed deposit change the price offered to the next identical deposit. If the selected financial mechanism requires solving a batch-wide allocation equation, that solution and its tie-breaking rules must be specified rather than hidden behind arbitrary transaction order.

Publication should be atomic from the user's economic perspective. A partially processed batch must not expose updated values to one group while leaving others at stale values. The engineering design must choose atomic commitment or a resumable process whose intermediate states cannot be traded or redeemed inconsistently.

The one-minute cycle is the normal path for both entry and exit. A prolonged oracle or network failure invokes the published failure policy instead of creating a fictional valid minute.

8. Binding requests and the timing boundary

Next-publication execution protects existing holders only when execution uses the new valuation. Delaying an order while preserving an old price would leave the same stale-entry opportunity.

Event	Economic treatment
Asset outperforms before publication	Existing holders receive the interval's revaluation
New allocation is queued	It waits without acquiring the preceding interval's gain
Snapshot is accepted	Existing claim values update first
New allocation executes	Units are acquired at the updated claim value
Redemption executes	Units are burned at that same updated valuation basis

Suppose a user sees BTC rise during the current minute. Buying a BTC allocation before publication must not let the user enter at the previous minute's claim value. Equally, a holder who sees BTC underperform cannot avoid the applicable revaluation by requesting redemption against that previous value.

Requests need a binding point before their execution valuation becomes available. Cancellation after learning whether a fill is favorable would create a free option against the pool. Users may instead submit conditions in advance, such as an expiry batch or minimum acceptable proceeds. Those conditions are evaluated mechanically and the treatment of an unfilled request is known beforehand.

Source units and pending cash must be reserved so they cannot support conflicting requests. A switch has one source and one destination; its source remains exposed until the batch revaluation, and its destination begins exposure after execution. Repeated contradictory instructions must have a defined replacement or rejection rule.

Batch processing has precedent in market-design research. Budish, Cramton and Shim analyze frequent uniform-price auctions as a response to speed competition. COX's proposed oracle-based pool processing is not that auction, and their findings do not prove COX's security. The shared lesson is that the timing and grouping of execution are economic design choices. [5]

The intended protection is narrow and testable: a user cannot acquire a prior interval's gain or avoid its loss by receiving a stale fill. It does not establish immunity to front-running. Selective inclusion, censorship, advance order knowledge, price-source manipulation and predictable sampling remain relevant threats.

The paper therefore uses "next-publication pricing" as the product rule. It does not claim that merely waiting one minute makes every adversarial strategy inefficient.

9. The closed collateral system

Let C_k be active collateral backing participating claims after publication k . Let $V_{i,k}$ be the aggregate redeemable value of claim class i . In the simplest no-fee accounting model:

$$V_{i,k} \geq 0 \quad \sum_i V_{i,k} = C_k$$

During revaluation alone, C does not change. Gains and losses redistribute its ownership. External deposits increase backing when activated; executed redemptions reduce it. Fees, if introduced, require a named recipient and explicit accounting rather than an unexplained discrepancy.

All accounting is measured in units of the collateral asset. Dollar examples assume, solely for illustration, that one collateral unit equals one dollar. Full backing in collateral units does not guarantee dollar purchasing power if the collateral itself changes in value.

ILLUSTRATIVE REDISTRIBUTION - TOTAL BACKING IS UNCHANGED

Before	A: 100	B: 50	150
After	A: 90	B: 60	150

For example, a pool with claims worth 100 and 50 may reallocate them to 90 and 60 under an agreed rule. Total backing remains 150. If the second holder redeems 60, the pool retains 90 for the first holder. These amounts illustrate conservation; they are not derived from any particular price movement.

Closed means no outside party is required to manufacture the winner's gain. It does not mean the pool size remains fixed forever, that holders cannot lose all their allocation, or that an arbitrary return promise becomes affordable. There is no borrowing or negative claim balance in the intended model.

Conservation and redeemability are related but separate. A fully backed claim still needs liquid, accessible collateral and a functioning execution path. The product's aim is direct redemption through valid publications, without requiring a secondary buyer. Achieving that aim requires proving the mechanism under mass redemption, not only under ordinary trading. [1, sections 30-31]

The backing invariant must cover custody obligations as well as active positions. Pending deposits belong to their depositors until activation; settled-but-unclaimed withdrawals remain liabilities. Neither category is spare capital available for performance redistribution.

The final mechanism must state whether operational reserves exist, who owns rounding residuals and how fees affect displayed returns. It must be possible for an independent observer to reconcile assets with every category of claim.

10. Why a COX reference is not a payout formula

An asset's reference can rise while its holder's monetary return depends on allocation and entry state. This distinction is foundational to the RPM framework. If most collateral is already positioned in the outperformer, little opposing collateral may be available to fund gains. [1, sections 10-11]

Consider the following illustrative revaluation, restricted to positive prices and positive total weighting:

$$V'_i = C (V_i h_i) / \sum_j (V_j h_j) \quad h_i = (P'_i / P_i) / (B' / B)$$

The sum of the new values equals C. However, the common benchmark factor cancels:

$$V'_i = C V_i (P'_i / P_i) / \sum_j [V_j (P'_j / P_j)]$$

Under this particular rule, asset claims compete according to asset returns and existing allocations. CRYPTO remains a reporting reference; changing its return alone does not alter dollar redistribution. This is not a proof that every closed-pool mechanism must behave this way. It is a reason not to present the first normalization formula as the completed product.

Boundary case	Required interpretation
All active collateral selects one asset	Revaluation cannot increase the pool's total backing
All tradable assets beat a broader CRYPTO basket	All references may rise; all claims cannot gain dollars together without a funding allocation
One asset is heavily crowded	Correct relative direction need not imply a large monetary gain
No positive weighted claims remain	The illustrative formula is undefined; explicit terminal handling is required

Adding a CRYPTO position with separately defined benchmark-growth participation can introduce an economic comparison side. It does not create an unlimited reserve or guarantee exact relative returns for every allocation. If nobody owns the funding side, a mechanism must expose that economic limit rather than silently assign it to pending deposits or an operator.

This is the main unresolved financial question for the subsequent mechanism specification: what transfer rule best expresses asset-versus-CRYPTO views while preserving nonnegative claims, fair entry and exit, and useful behavior under extreme allocations?

The product paper specifies the desired experience and the constraints that a candidate must satisfy. It does not select the illustrative rule above as the production mechanism. A simple conservation proof is necessary but insufficient evidence of economic fairness, manipulation resistance or faithful benchmark exposure.

11. Continuous entry, switching and redemption

A claim-unit model can express the accounting requirements without pretending to solve the transfer rule. Let U_i be outstanding units in class i . After the old positions have been revalued, let p_i be the class's current redeemable value per unit:

$$p_i = V_i / U_i \quad \text{for } U_i > 0$$

At a positive p_i , a deposit d allocated to that class receives d/p_i units. A redemption of x units receives $x p_i$ collateral. The supply changes with the corresponding backing. These identities prevent a new participant from entering at an obsolete unit value.

$$\text{Minted units} = d / p_i \quad \text{Redemption proceeds} = x p_i$$

A switch burns units in its source class and uses the resulting current value to acquire units in its destination class. It is not a second opportunity to apply the interval's price return. Fees and rounding, if any, are deducted according to the same published rules for every participant.

All flows use the batch's consistent post-revaluation, pre-flow valuation basis unless the final mechanism explicitly requires a different simultaneous solution. Processing order must not create different economic outcomes for equivalent requests. Capacity limits, if present, require deterministic allocation and rejection rules.

Pending deposits are segregated from active backing. Existing positions submitted for redemption remain exposed until their applicable revaluation; afterward, executed withdrawal amounts are reserved and no longer participate in future performance. The custody ledger must account for active backing, pending deposits, withdrawal payables and separately owned reserves or fees.

Several boundary cases cannot be hidden inside the unit-price equation. A class with no units needs a bootstrap rule. A class with zero value cannot accept deposits through division by zero. A fully empty pool needs a restart procedure. A token reaching a valid zero price needs a terminal-loss convention before any subsequent ratio is evaluated. Integer rounding must not reward splitting one order into many small orders.

The required proof covers arbitrary sequences: revaluation, entry, partial exit, complete exit, switches and repeated re-entry. The pool must preserve remaining holders' backing after each operation, including when all profitable holders redeem in the same batch.

Continuous access is therefore a property of the whole accounting transition, not merely the existence of deposit and withdrawal buttons.

12. A worked publication and user journey

Consider a synthetic pool with two claim classes. Before the next publication, A has 100 units at 1 collateral unit each and B has 50 units at 1 each. Active backing is 150. A new depositor has separately escrowed 30 for class A. A B holder has requested redemption of 10 B units.

Assume the eventual transfer rule assigns 90 to A and 60 to B at this publication. This assumption isolates entry and exit accounting; it is not a claimed derivation from asset prices.

Stage	A backing / units	B backing / units	Other custody obligations
Before revaluation	100 / 100	50 / 50	30 pending deposit
After revaluation	90 / 100	60 / 50	30 pending deposit
Activate deposit	120 / 133.333...	60 / 50	None from deposit
Execute redemption	120 / 133.333...	48 / 40	12 withdrawal payable
Pay withdrawal	120 / 133.333...	48 / 40	None

After revaluation, A's unit value is 0.90 and B's is 1.20. The new 30 deposit receives 33.333... A units at 0.90. The B redemption burns 10 units and receives 12. Remaining active backing is 168, equal to A's 120 plus B's 48. Total collateral paid in was 180; 12 has left. No money was created.

The repeating decimal is exact rational arithmetic in this example. A real implementation must apply a declared integer-unit and residual policy, then verify the same conservation relationship at its chosen precision.

For the user, the journey is simpler than the ledger. They choose an asset, inspect its current COX reference and position terms, enter an amount, review the next eligible publication and submit. The interface shows the request as queued rather than filled. After commitment, it displays issued units, actual valuation, fees and publication identifier.

A holder who rotates follows the same path: the current allocation remains active through revaluation, then the switch executes. A holder who exits sees estimated proceeds before commitment and final proceeds after execution. An estimate must not masquerade as a guaranteed stale quote.

The portfolio distinguishes deposited amount, current redeemable value, realized withdrawals and pending requests. COX reference change is displayed separately from personal profit and loss. A user should never have to infer their account return from the public reference chart alone.

13. Product surfaces, candles and expectations

Each asset page should explain three quantities: the underlying reference price, its COX reference against CRYPTO, and the current value of the user's claim. They answer different questions and can move differently because the financial mechanism incorporates pool participation.

Surface	What it communicates
Asset page	Price performance, COX reference, latest accepted time and next publication
CRYPTO page	Composition, weights, index history and constituent contributions
Portfolio	Units, redeemable value, deposited basis, realized flows and queued instructions
Publication history	Prices, methodology, batch identity and reconciliation outcome
System status	Fresh, queued, delayed, rejected or recovery state

One-minute publication does not automatically produce a conventional intraminute trading candle. If there is one accepted reference sample each minute, the product has one reference observation per minute. It cannot infer unseen highs and lows between those samples.

Longer-period reference candles can aggregate accepted observations, with their sampling convention disclosed. Executed-price candles require actual fills. An indicative spot-price chart may update more frequently, but it must be labeled separately and must not imply that users can execute against those indications. Missing publications remain visible as gaps or stale states rather than synthetic market activity.

An explanation of a COX move should distinguish an asset move from a benchmark move. "BTC rose, but CRYPTO rose faster" is a complete relative-performance explanation. A claim-value explanation may additionally need to identify allocation effects under the selected transfer rule.

The EOX paper describes a distinct market expectation price for a future terminal reference. COX's current design does not automatically inherit that object. If the protocol issues and redeems units at calculated values, a separate freely floating expectation price requires its own market mechanism. Expectations may instead be expressed through how users allocate collateral and how crowded each class becomes.

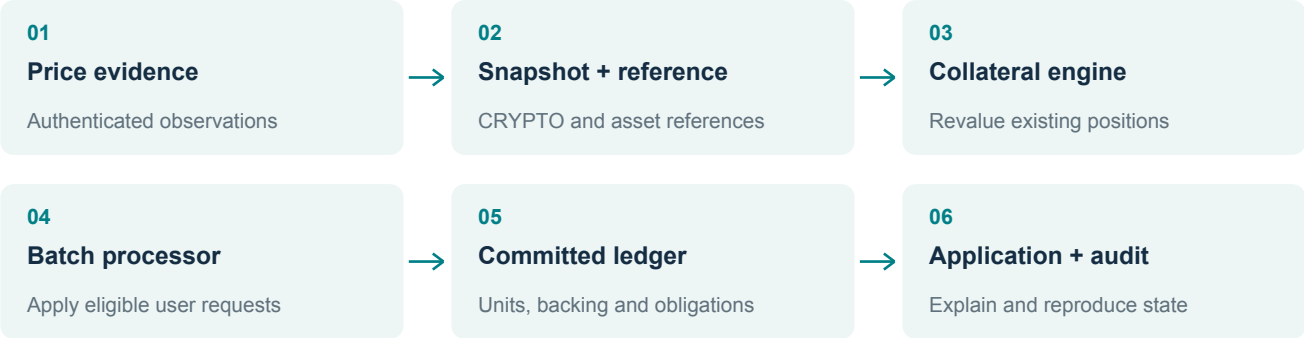
The interface should therefore avoid labeling a calculated redeemable value as an implied forecast. An optional secondary market would need to explain premiums, discounts and the interaction with issuance and redemption. It is not assumed in this version.

The product remains continuous without a mandatory terminal epoch. Methodology versions and observation origins provide historical organization; they are not expiry dates for every position.

14. System architecture and persistent state

COX separates the external price world from the deterministic reference and collateral transitions. The architecture should be independently reproducible at each boundary.

READ LEFT TO RIGHT, THEN CONTINUE ON THE SECOND ROW



The source adapter authenticates observations and enforces their schema. The snapshot builder selects a synchronized admissible set under the pinned observation rule. The reference engine computes CRYPTO and asset references. The collateral engine applies the selected transfer rule to existing claims. The batch processor then applies eligible user flows, and the ledger commits the reconciled state.

The application reads committed state for current claim valuations and labels pending or indicative information. A queued request executes at its assigned batch's future valuation; the latest displayed value is only an estimate for that request. It must not calculate an alternative personal version of CRYPTO or substitute a spot price for a claim value.

Persistent object	Essential purpose
Methodology version	Pin sources, weights, selection rules, arithmetic and activation
Price snapshot	Bind accepted observations and their provenance
Publication	Link prior state, references, batch and resulting ledger
User request	Bind owner, operation, batch, reserved value and prior conditions
Claim ledger	Track units, class backing and all custody obligations
Execution receipt	Explain fills, rejections, fees and resulting balances

A publication must identify its predecessor and cannot be replayed as a new interval. Recovery after a crash must resume or reject an interrupted operation deterministically; it must never apply the same price movement or withdrawal twice.

The chain, storage technology and oracle transport are intentionally unspecified here. Choosing them does not determine the economics. A production implementation must nevertheless define its finality boundary, authentication model, operator permissions, source licensing and archival availability.

EOX components may provide useful patterns for immutable evidence, versioned interfaces and reproducible calculations. Reuse requires review: country identifiers, economic normalization, long challenge windows and annual settlement assumptions cannot simply be relabeled as crypto logic.

This paper is a separate product design. It does not modify the EOX engineering contract, assign developers new delivery obligations or declare the existing EOX infrastructure to be a deployed COX system.

15. Security, failure and governance

COX's principal risks follow from executable prices and conserved claims. The security objective is to prevent a participant from extracting value through distorted inputs, privileged timing or inconsistent accounting rather than through the specified economic outcome.

Threat	Required design response
Stale-price entry or exit	Revalue first; execute at the applicable new valuation
Favorable timestamp selection	Deterministic sampling and synchronized admissibility
Price manipulation	Robust source rules and economically justified exposure bounds
Cancellation after learning value	Binding requests or conditions fixed in advance
Selective batch inclusion	Auditable admission, cutoff and execution rules
Duplicate or partial processing	Unique identities, atomicity and deterministic recovery
Rounding extraction	Declared precision and residual rules; split-order testing
Benchmark governance abuse	Prospective versions, continuity and visible activation

The value extractable from corrupting a price must be assessed across every affected claim. A constituent can influence its own reference, CRYPTO and other references. Authentication verifies origin; it does not establish an unlimited manipulation budget.

If the required snapshot is invalid, the normal executable batch is delayed or rejected according to a published rule. The interface preserves the last accepted reference with its age and does not silently trade at it. The engineering specification must decide whether a failed constituent halts the common pool or invokes a previously defined fallback; removing it opportunistically would change the benchmark.

Queued orders need an expiry and recovery policy. New deposits awaiting activation remain attributable to their owners. Existing claim holders cannot automatically reclaim their original deposits after losses merely because processing is paused. A prolonged outage needs a predetermined unwind or recovery path that does not reward the first user to escape.

Governance may update future methodology, maintain sources and invoke bounded emergency procedures. It must not arbitrarily set COX values, rewrite settled withdrawals or alter a live batch to favor a participant. Version activation without terminal epochs needs notice, a clear transition and preservation of existing monetary rights.

Collateral custody and denomination remain material assumptions. Conservation in units does not protect against a stolen reserve or loss of the collateral's value. Operational solvency therefore requires both correct accounting and control over the backing it describes.

The term RPM specifies an economic design, not a legal exemption. Legal classification and permitted operation depend on the eventual claim, custody and jurisdiction and remain outside this product paper's conclusions. [1, section 38]

16. What changes from EOX Paper 3

COX keeps the common benchmark, reproducible references and closed-pool ambition. It changes the object measured and the timing of participation. Those changes require a distinct product paper.

EOX concept	COX treatment
Multidimensional economic engine	Normalized reference prices and a price index
Interpretation of growth, inflation and labor	Explicit definition of price return
WORLD from normalized country states	CRYPTO from an eligible asset universe
Statistical sources and vintages	Price observations, feed identity and observation time
Confidence factors around economic evidence	Input-quality admission without return attenuation
Optimistic evidence challenge path	Authenticated inputs and deterministic checks initially
Optional signal amplification	1x relative price expression
Sparse economic events	Scheduled one-minute price publications
Finite market epochs	Continuous claims and next-publication flows
Terminal expectation price	Allocation expresses outlook; independent price is optional
Closed collateral as research direction	Same requirement, with explicit batch-accounting boundaries

The measurement burden falls substantially. COX need not establish a universal interpretation of heterogeneous economic indicators. Its assumption is narrower: price performance is the phenomenon users want to compare.

The timing burden rises. Public prices can change before the system commits its next state, so every entry, exit and switch needs an unambiguous valuation boundary. One-minute publication organizes this boundary but cannot substitute for correct pricing rules.

The benchmark question remains central. "Crypto performed well" depends on which assets count and how they are weighted. Excluding market cap and volume from performance does not eliminate the need for that specification.

The financial burden remains. Closed collateral has to support fair continuous entry and exit under uneven allocations, failures and mass redemption. Paper 3 can establish required behavior and reject misleading formulas. It cannot convert an illustrative normalization identity into a production proof.

The strongest distinctive claim for COX is therefore practical and testable: users can express relative price views through a transparent, continuously accessible, fully backed allocation system whose references and monetary rights can both be independently reconstructed. Whether a final mechanism meets that claim is the next research stage.

17. Validation program and remaining decisions

The next stage is mechanism derivation followed by simulation. No backtest, user study or production security result is claimed in this paper. One minute is the selected initial cadence to evaluate.

Experiment	Required result
Equal asset and benchmark returns	Ratio-relative reference remains unchanged
Common quote-currency rescaling	Matched ratios remain consistent
Constituent rebalance	No artificial return at the transition
One-sided and highly crowded pools	No manufactured backing or hidden funding source
Deposit after an observed gain	New units cannot acquire prior performance
Redemption after an observed loss	Old holders cannot escape at a stale value
Equivalent batch permutations	No unfair advantage from processing order
Split orders and integer rounding	No extractable value from fragmentation
Mass exit and complete pool drain	All remaining liabilities stay backed
Missing prices and interrupted execution	No stale fills, replay or partial-state exploitation
Zero prices and empty classes	Defined terminal and restart behavior
Adversarial source or order timing	Quantified limits rather than a blanket immunity claim

Mechanism research must compare candidate transfer rules on more than their sum. It should test whether the claim responds intelligibly to asset-versus-CRYPTO performance, how crowding changes outcomes, whether identical terminal prices reached by different paths yield different wealth, and whether repeated switching can extract collateral without taking the intended exposure.

The minimum deliverables are a benchmark specification, a transfer and unit-accounting specification, an order and publication state machine, and reproducible adversarial simulations. A later engineering paper should provide exact arithmetic and proofs; a security paper should attack their assumptions and report remaining failure modes.

The open choices are concrete: final assets and identities; weight schedule; source and sampling rules; holdable CRYPTO rights; redistribution and fee functions; collateral and custody; rounding and empty-state treatment; delay and emergency policies; and governance activation. Those choices do not reopen the price-only, one-minute and continuous-pool direction established here.

COX succeeds as a product concept if users can understand what they are measuring and what they own. It succeeds as a financial system only if the mechanism honors that explanation through every valid transition. Price evidence, a relative reference and a solvent claim are connected objects, but each needs its own exact definition.

Appendix A. Canonical terminology

Term	Meaning in this paper
COX	Crypto Outlook Index
Reference price	The accepted asset price under the configured source and timing rules
Price performance	Change in reference price over a defined common interval
CRYPTO	The shared benchmark index, with a versioned universe and weights
COX reference	Normalized asset price divided by normalized CRYPTO, displayed from 100
Relative change	Ratio of gross asset and benchmark returns, minus one
Claim class	A category of pool rights with a defined revaluation rule
Claim unit	An accounting unit representing a share of a claim class's backing
Redeemable value	Collateral attributable to a claim under the current committed state
Active backing	Collateral participating in performance redistribution
Pending deposit	User collateral awaiting activation, outside prior-interval performance
Publication	A committed reference and ledger transition for an identified batch
Batch	Requests admitted before a defined cutoff and processed together
Revaluation	Redistribution among existing claims before new flows execute
Continuous participation	Requests can be submitted throughout operation; fills use valid publications
Quality admission	Acceptance or rejection of observations under objective input rules
Closed collateral	Revaluation cannot create claims exceeding the backing available to them

Interpretive limits. "Fully backed" describes the required relationship between claims and collateral; this draft is not a certification of implemented solvency. "Global CRYPTO" must be qualified by actual benchmark coverage. "No leverage" does not mean no loss. "Next publication" means the next applicable valid publication after admission, with failures governed by an explicit policy. "Price only" excludes non-price performance scores while retaining the source metadata needed to verify prices.

Document lineage. Paper 1 supplies the general RPM framework. EOX Paper 3 supplies the product-layer precedent and the separation of evidence, reference and claims. COX Paper 3 adapts that framework to the price-only product direction developed on 10 October 2026. It is a standalone draft and does not supersede EOX's engineering contract.

References

[1] Relative Performance Markets: Extending Prediction Markets and Derivatives Through Comparative Outcome Trading. Project research manuscript, Paper 1. Local source: docs/papers/relative_performance_markets_paper_1.pdf. Especially sections 9-15, 25-32 and 34-38. Foundational framework; not a peer-reviewed validation of the proposed COX mechanism.

[2] EOX: Economic Outlook Index - A System for Measuring and Trading Economic Performance Relative to the World. Product and System Paper, version 0.1, September 2026. Local source: docs/papers/eox_paper_3.pdf. Especially sections 4-9, 12-24 and 40-43.

[3] Pyth Network. Best Practices, Pyth Core Developer Documentation. <https://docs.pyth.network/price-feeds/core/best-practices>. Accessed 10 October 2026. Used for examples of oracle staleness, latency, adversarial update selection and uncertainty; no vendor selection or COX security proof is implied.

[4] S&P Dow Jones Indices. FAQ: S&P Cryptocurrency Index Series. <https://www.spglobal.com/spdji/en/education/article/faq-sp-cryptocurrency-index-series/>. Accessed 10 October 2026. Used as precedent for explicit crypto index eligibility and reference methodology, not as the COX benchmark.

[5] Budish, E., Cramton, P., and Shim, J. (2015). The High-Frequency Trading Arms Race: Frequent Batch Auctions as a Market Design Response. The Quarterly Journal of Economics, 130(4), 1547-1621. <https://doi.org/10.1093/qje/qjv027>. The paper studies a distinct auction design; its results are not asserted to hold for COX's proposed pool processing.

Source and evidence note. The product commitments in this document originate in the COX design discussion. The algebraic examples are derived within the paper. External references provide limited context for index design, oracle integration and batching. No historical return dataset, calibration, production benchmark or empirical manipulation test was used to establish the proposed mechanism.